



A Richards growth model to predict fruit weight

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Summary

The Richards model comprises several popular sigmoidal and monomolecular growth curves. We illustrate fitting of a Bayesian Richards model by splitting the full growth model into several submodels, followed by a model selection procedure. The performance of the methodology is evaluated by Monte Carlo simulations. A double-sigmoidal version of the Richards model is applied to model grape bunch weight based on data from a New Zealand vineyard over a single growing period.

Key words: growth model; model selection; Markov chain Monte Carlo; yield prediction

1. Introduction

Growth models are a useful tool to describe the change in size and weight of crops over time. Given some prior knowledge about the location of the growth model parameters, and samples of measurements at early stages of the growth cycle, the expected final size at harvest time can be predicted. This yield prediction can be used for planning of resources and crop management. The growth curve can give insight into the physiological development of a plant, for example, helping with the identification of different growth phases or the response to external factors such as climate change.

Here, we are focussing on modelling fruit weight; especially, looking at the growth of fleshy fruits and berries. The growth of grape berries can be described by two sigmoidal cycles (Coombe & McCarthy 2000), that are joined together to form a double sigmoidal curve. Fernandes, Pereira & Muniz (2017) have tested several nonlinear models to describe such a curve for the growth of coffee berries. They have considered single logistic, double logistic, single Gompertz and double Gompertz curves, and found the double logistic and double Gompertz to be good models, with the former slightly superior to the latter. These models are relatively simple, with a small number of parameters, avoiding estimation problems for small sample sizes. The choice of a simple growth curve comes at a cost of less flexibility and requires strict assumptions about the shape of the curves; for example, the logistic curve is symmetric around the inflection point.

In order to account for rapid growth in the first phase of the curve, Genard *et al.* (1999); Dai *et al.* (2009) used a combination of monomolecular and logistic models to describe change in peach and grape berry diameter, respectively. They used least-squares

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minimisation to find estimates of their six model parameters. These models provided a good fit, especially when there are no observations available early on, at the start of the growing period.

Richards (1959) described the properties of an extended form of the von Bertalanffy's growth model, which facilitates the comparison between sigmoidal growth curves of quite different shapes. The Richards curve includes several popular models such as the logistic model and the Gompertz model as special cases; hence, it is a very flexible tool. The Richards curve has been successfully applied, for example, to buckwheat plant growth modelling (Gregorczyk 1998).

In this paper, we implement the Richards model within a Bayesian framework. Instead of estimating the posterior distribution for the parameter governing the family of the fitted curve, we decompose the parameter space into two mutually exclusive and exhaustive subspaces, fit a separate model for each, and use predictive stacking Yao *et al.* (2018) to evaluate their respective weights. One can then either choose the model with the highest weight or do Bayesian model averaging. We demonstrate the method using the grape bunch growth data collected from the Rowley Crescent Vineyard in Marlborough, New Zealand.

2. The Richards growth model

The Richards model can be written as a differential equation (Seber & Wild 2003)

$$\frac{df}{dx} = \frac{\kappa}{1-\delta} f \left[\left(\frac{f}{\alpha} \right)^{\delta-1} - 1 \right],$$

with the solution

$$f(x) = \alpha \left[1 + (\delta - 1) \exp(-\kappa(x - \gamma)) \right]^{\frac{1}{1-\delta}},$$

where $\delta \neq 1$ and for $\delta < 1$ the restriction $(1 - \delta) \exp(\kappa\gamma) \leq 1$ is required to ensure that $0 \leq f \leq \alpha$ (Jorgensen 1981). The parameters of the model have a physical interpretation; α denotes the final size, the coordinates of the point of inflection are determined by $x = \gamma$ and $f = \alpha\delta^{\frac{1}{1-\delta}}$, and κ acts as a scale parameter on x . It is easy to see that this parameterisation includes several popular growth curves, with the monomolecular model at $\delta = 0$, the logistic model at $\delta = 2$, the von Bertalanffy model at $\delta = 2/3$, and the Gompertz model at $\lim \delta \rightarrow 1$. An overview of different model parametrisations and parameter interpretation can be found in Tjørve & Tjørve (2010).

There is a clear difference in curves for $\delta > 1$ compared to $-1 < \delta < 1$, where in the latter case the inflection point is moved to 0; then, instead of a sigmoidal shape, we end up with monomolecular-type models where the exponential increase of the curve up to the inflection point is omitted, resulting in an abrupt start of the growth curve with a decreasing gradient up to the final size.

With the restriction of $\delta \neq 1$, we can separately look at a set of functions with $\delta > 1$, and a second set with $-1 < \delta < 1$

$$\begin{aligned} f_1(x; \alpha_1, \delta_1, \kappa_1, \gamma_1) &= \alpha_1 \left[1 + (\delta_1 - 1) \exp(-\kappa_1(x - \gamma_1)) \right]^{\frac{1}{1-\delta_1}} & \text{for } \delta_1 > 1, \\ f_2(x; \alpha_2, \delta_2, \kappa_2, \gamma_2) &= \begin{cases} \alpha_2 \left[1 + (\delta_2 - 1) \exp(-\kappa_2(x - \gamma_2)) \right]^{\frac{1}{1-\delta_2}} & \text{for } -1 < \delta_2 < 1 \wedge x > \gamma_2, \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

For both curves, we assume the same interpretation for the final size α_1 and α_2 , but with a different range for the parameters δ_1 and δ_2 , we assume that also the parameters κ_1 , κ_2 and γ_1 , γ_2 will take on different values.

Given a sample (y_i, x_i) with $i = 1, 2, \dots, n$, we assume the Bayesian models (Gelman *et al.* 2013):

$$y_i | x_i, \alpha_1, \delta_1, \kappa_1, \gamma_1 \sim N(f_1(x_i | \cdot), \sigma^2),$$

$$y_i | x_i, \alpha_2, \delta_2, \kappa_2, \gamma_2 \sim N(f_2(x_i | \cdot), \sigma^2).$$

For the Richards curve parameters, the following prior distributions are specified:

$$\begin{aligned} \alpha_1 &\sim N(\mu_{\alpha_1}, \sigma_{\alpha_1}^2)_0^\infty, & \kappa_1 &\sim N(\mu_{\kappa_1}, \sigma_{\kappa_1}^2)_0^\infty, & \gamma_1 &\sim N(\mu_{\gamma_1}, \sigma_{\gamma_1}^2)_0^\infty, \\ \alpha_2 &\sim N(\mu_{\alpha_2}, \sigma_{\alpha_2}^2)_0^\infty, & \kappa_2 &\sim N(\mu_{\kappa_2}, \sigma_{\kappa_2}^2)_0^\infty, & \gamma_2 &\sim N(\mu_{\gamma_2}, \sigma_{\gamma_2}^2)_0^\infty, \end{aligned}$$

$$\delta_1 \sim N(\mu_{\delta_1}, \sigma_{\delta_1}^2)_1^\infty, \quad \delta_2 \sim N(\mu_{\delta_2}, \sigma_{\delta_2}^2)_{-1}^1,$$

The notation $N(\cdot)_a^b$ defines a truncated normal distribution with lower bound a and upper bound b . Assuming truncated prior distributions allows us to ensure strictly increasing growth, and to restrict the parameters to their designated subspace.

For the residual variance, we assume

$$\sigma^{-2} \sim \text{Gamma}(0.001, 0.001).$$

Samples from the posterior distribution of the parameters are obtained by a Metropolis–Hastings algorithm (Metropolis *et al.* 1953; Hastings 1970), using Gibbs sampling (Geman & Geman 1984) for the residual variance.

3. Selecting curves and model averaging

Instead of directly evaluating the posterior distribution of the parameter δ in a range of $\delta \in (-1, 1) \cup \delta \in (1, \infty)$, the estimation problem is decomposed into fitting the two models separately and then selecting the model that provides a better prediction for new data.

The selection of the model with the best predictive performance out of a set of two or more models, requires the definition of a selection criterion or a loss function from which an estimate for a model weight can be derived (Claeskens & Hjort 2008).

In order to estimate the model weights for our set of Bayesian Richards models, we are using predictive stacking, following Yao *et al.* (2018). The prediction accuracy is assessed by exact leave-one-out cross-validation: re-fitting the Bayesian model, leaving out each of the $i = 1, \dots, n$ observations one at a time. The stacking weights for $k = 1, \dots, K$ models M_k are obtained by solving the optimisation problem

$$\max_{\omega} \frac{1}{n} \sum_{i=1}^n \log \sum_{k=1}^K \omega_k p(y_i | y_{-i}, M_k).$$

Here, $p(y_i|y_{-i}, M_k)$ denotes the posterior predictive density at the i th left-out observation conditional on the set of remaining observations with

$$p(y_i|y_{-i}, M_k) = \int p(y_i|\theta_k, M_k)p(\theta_k|y_{-i}, M_k)d\theta_k$$

for $\theta_k = (\alpha_k, \delta_k, \kappa_k, \gamma_k, \sigma_k)^\top$. The ‘best’ model is selected as the model with the largest estimated weight. The estimated weights can also be used to calculate weighted averages of pointwise predictions.

Exact leave-one-out cross-validation will increase the computation time, especially for large data sets; also the number of models in the set cannot be too large, as each model needs to be refitted n times to compute the corresponding weights.

For the derivation of the stacking procedure, combining a set of Bayesian posterior predictive distributions, and approximations to the leave-one-out predictive density, we refer the interested reader to Yao *et al.* (2018) and Vehtari, Gelman & Gabry (2017).

4. Simulation study

For a range of different parameters δ , we are estimating the frequentist probability of selecting a specific model, where the set of models consists of the two Richards curves f_1 and f_2 , and a third curve, where δ is fixed at 2, which results in a logistic growth model. The true growth curves with different values for parameter δ are shown in Figure 1.

We have chosen arbitrary values for the true parameter settings of $\alpha=5$, $\kappa=1.5$, $\gamma=5$, $\sigma=0.2$, generating $n=10$ observations for each iteration of the Monte Carlo simulation with $x_i \sim U(0, 10)$.

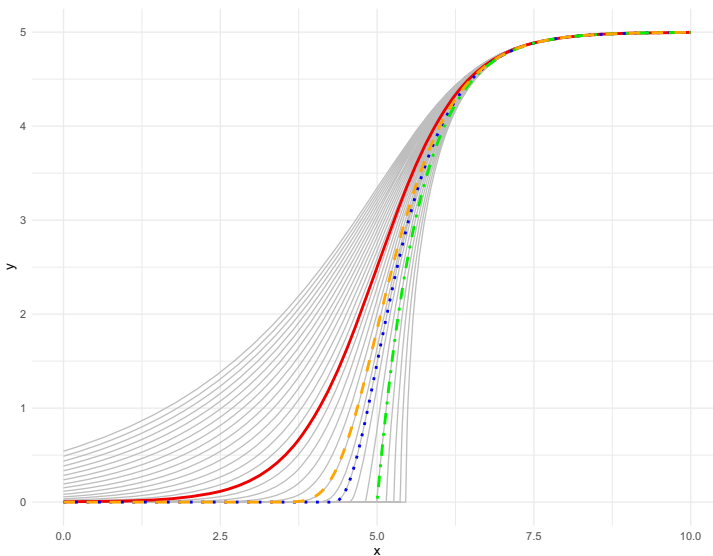


Figure 1. Parameter settings for the simulation study. Each curve represents a different value for the parameter δ ; the special cases (from left to right) of the logistic (solid), Gompertz (dashed), vonBertalanffy (dotted), and monomolecular (dash-dot) model are highlighted.

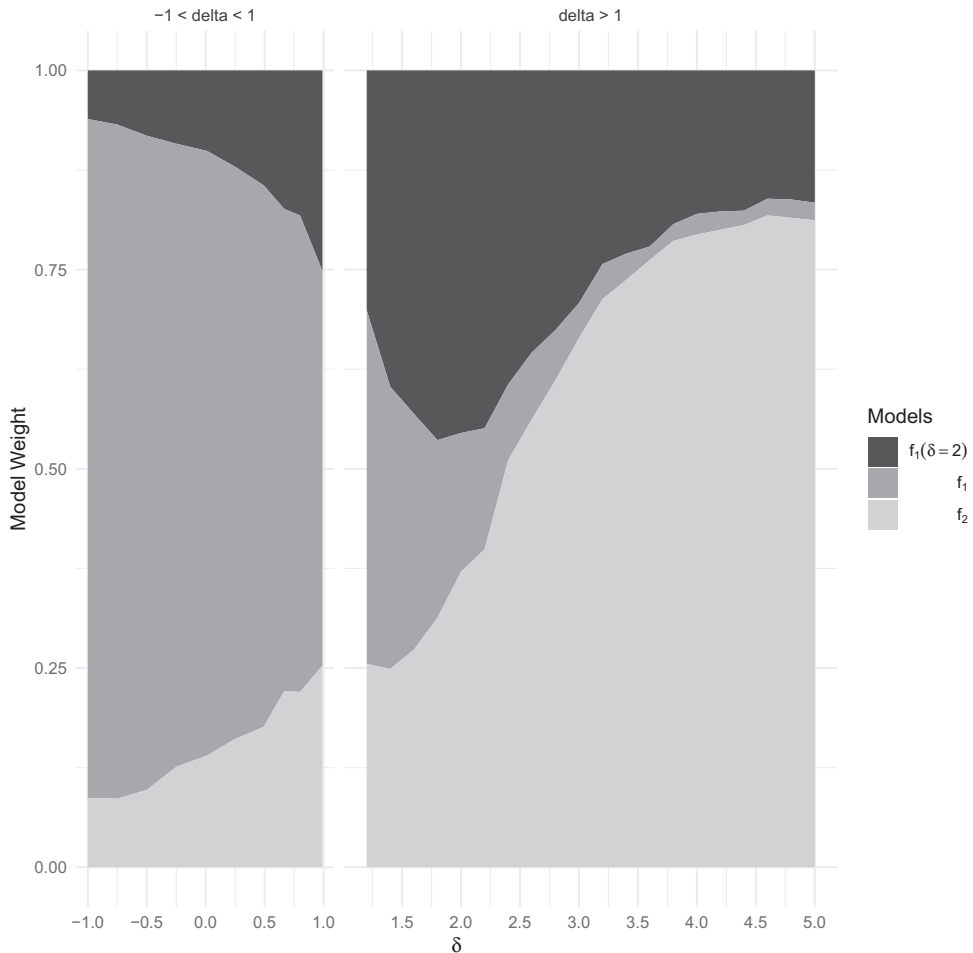


Figure 2. Simulation results showing the estimated probability of selecting one of three models ($\delta=2$, $-1 < \delta < 1$, $\delta > 1$) as the ‘best’ model with the highest prediction accuracy for new data.

For each parameter in the model a weakly informative prior is assumed, obtaining 1,000 samples from the posterior distribution after initial 9,000 iterations of a burn-in phase. For each value of δ , 1,000 simulation runs are performed, selecting the ‘best’ model from each run based on the estimated predictive stacking weights. We report the proportion of models selected as ‘best’ given each value of δ , shown in Figure 2.

For the chosen simulation settings, predictive stacking enables us to distinguish reasonably well between the $-1 < \delta < 1$ and $\delta > 1$ scenarios. For $\delta=2$ there is still a large probability to select the $\delta > 1$ model as $\delta=2$ is a subset of its model space. With a smaller distance between the two models, they only differ in terms of complexity.

5. A double Richards curve for modelling fruit growth

The growth of many fleshy fruit can be described in three stages: two sigmoidal growth cycles divided by a lag phase or rest period. With the assumption of a Richards function

for each of the two sigmoidal patterns of fruit growth, we can create a double sigmoidal function by the sum of both curves as a function of time,

$$f_{j,k}(t; \cdot) = f_j(t; \cdot) + f_k(t; \cdot),$$

where $f_j(t; \cdot)$ with $j = 1, 2$ and $f_k(t; \cdot)$ with $k = 1, 2$ is the Richards curve representing the first and second growth phase, respectively.

Some constraints have to be incorporated in the model, which will allow us to uniquely identify each of the Richards curve components with one of the growth phases. We redefine some parameters as differences and use a fixed boundary of 0. Hence, α_k in the second growth phase represents the distance to α_j in the first growth phase, and γ_k represents the distance to γ_j . Both parameters α_k and γ_k have a prior distribution truncated at 0. These parameter constraints are used to ensure identifiability of parameters, i.e. avoiding label switching of the first and second parameter for the inflection point with $(\gamma_k - \gamma_j) \geq 0$.

Defining the parameter δ in a range of $\delta \in (-1, 1) \cup (1, \infty)$ for the second part of the model as f_1 and f_2 , allows us to model growth after the rest period as a direct steep increase, which is sometimes observed in practice. The same applies for the first part of the curve when there is no information available in the early growth period.

A prediction for fruit weight can be obtained as the posterior mean of $f_{j,k}(t; \cdot)$ at a certain time t . Given the samples from the posterior distribution of $f_{j,k}(t; \cdot)$, credible intervals for the mean can be computed. If the uncertainty about the weight of an individual fruit is of primary interest, the credible intervals can be based on samples from the posterior predictive distribution.

6. Data example

Grape bunch weight was observed at the Rowley Crescent vineyard, Wairau Plains, Marlborough, New Zealand, sampling weekly over the growing period 2017 to 2018; that is, thirteen times in total with 15 or 16 observations each (Ellis *et al.* 2020). Only a subset of the data with apical bunches are used and a logarithmic transformation was applied to the observed weights. A scatterplot with the full data is shown in Figure 3. Instead of just predicting the final yield as described in Ellis *et al.* (2020), we are interested in selecting a model that represents the development of grape bunches from the start of the growing period until harvest. We are especially interested in modelling any abrupt change in growth directly after véraison (rest period).

Bunch weight was mostly sampled when reaching the plateaus at véraison (rest period) or near harvest. Not much information about the growth curve between the plateaus is available in the sample; then, most information about the growth depends on the provided prior information. We assume weakly informative prior distributions based on knowledge about phenological processes for grape growth in the region.

The set of models includes all combinations of parameter constraints $\delta > 1$, $-1 < \delta < 1$, and $\delta = 2$ for both phases of the double sigmoidal growth curve, resulting in a total of nine different models. Leave-one-out cross-validation was performed for each model, running each model for 100,000 iterations, and saving 1,000 samples from the posterior distribution. The estimated stacking weights are presented in Table 1.

There are four out of a set of nine models, which have a considerable weight. In the first phase of the grape growth the model selection procedure favours the $\delta > 1$ parametrisation.

Table 1. Posterior stacking model weights for each of nine different models with different constraints on the parameter δ in the two phases of the double sigmoidal growth curve.

Phase I	Phase II	Model weight
$\delta > 1$	$-1 < \delta < 1$	0.292
$\delta > 1$	$\delta = 2$	0.247
$\delta > 1$	$\delta > 1$	0.244
$-1 < \delta < 1$	$\delta > 1$	0.175
$-1 < \delta < 1$	$\delta = 2$	0.039
$\delta = 2$	$-1 < \delta < 1$	0.004
$-1 < \delta < 0$	$-1 < \delta < 1$	0.000
$\delta = 2$	$\delta > 1$	0.000
$\delta = 2$	$\delta = 2$	0.000

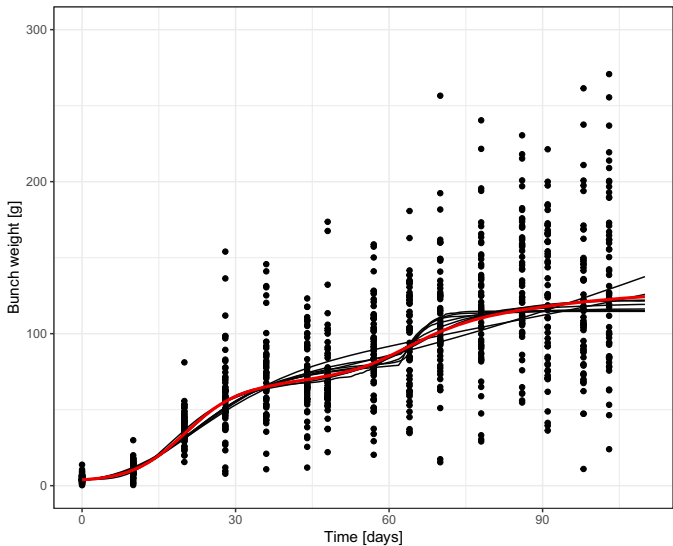


Figure 3. Observed bunch weights from the Rowley Crescent vineyard, and the predicted average weights for a set of nine different growth models with different constraints on the parameter δ . The highlighted line shows the model averaged prediction based on predictive stacking.

At the beginning of the growth period a clear exponential increase in bunch weight is visible, leading to the preference of models with $\delta > 1$; although, the change in weight is not symmetric around the inflection point as all $\delta = 2$ settings receive very low weights. For the second growth phase instead, not much data is available that is informative about the shape of the growth curve; hence, there is no clear decision about a specific trend, as models for any configuration about the parameter δ receive approximately the same weight.

The posterior median for the prediction of each separate model is shown in Figure 3 together with the weighted average prediction. It is worth noting that models with monomolecular growth in the first phase reach the final size at a much later time, exhibiting a less steep growth curve in the second phase.

7. Discussion

We have shown a flexible approach to model grape bunch weight by a combination of two Richards curves, allowing for asymmetry around inflection points and also the steep increase of a monomolecular growth model. Instead of estimating all parameters simultaneously and exploring the full parameter space of δ , the estimation problem is facilitated by splitting the full model into several submodels. Model selection by predictive stacking allows to choose the best submodel or average estimates given the model selection weights.

The growth model can easily be extended to include environmental variables, such as climate or spatial effects, to explain additional variability between the observed bunch weights. Any expert knowledge can be included through providing informative prior distributions.

As described in Section 2, each parameter has a physical/biological interpretation, i.e. final weight, location of the inflection point, and growth rate. This is an advantage compared to potentially a bit more flexible, non-parametric methods like regression splines, which only allow to make predictions. With splines, it is also more difficult to capture any abrupt change in growth as in a monomolecular parametrisation of the Richards model. The split into separate Richards models allows us to interpret the corresponding weights as probability for a certain class of growth model: monomolecular, logistic, a generalised logistic ($\delta > 1$), or choosing an even larger set of submodels.

With the estimated model selection weights, we are able to gain insight into the growth curve shape, and given that there is sufficient enough information available, we can connect features of the growth curve with phenological processes. The smaller complexity of the monomolecular curve can also be preferred, when there is not much information available at the start of the growth period; i.e. we might not be able to sample the weight of fruit right at the start of the season.

For the interpretation of model selection weights, we are restricted to look only at the point estimates. There are no uncertainty estimates available for these model probabilities. Also, any inference using model averaging is conditional on the point estimates of the weights. Given the difficulty of setting up a single full model, we do not think that the split into several submodels results in a large loss of information.

Of course, the computation time can become a problem for large datasets and/or large sets of models. The number of fitted models can be reduced by using K -fold cross-validation instead of choosing $K = n$ for leave-one-out cross-validation. For a nonlinear growth model, the order of data points can be of importance. In this case, random selection into K subsets is preferred. Vehtari, Gelman & Gabry (2017) discuss several methods to estimate the prediction accuracy from a fitted Bayesian model. They propose Pareto smoothed importance sampling to compute leave-one-out cross-validation estimates. Instead of fitting separate models for each of the K subsets, each model needs to be fitted only once. The importance sampling procedure has been shown to be a good approximation to the exact leave-one-out cross-validation.

Supporting and supplementary material

R code for the simulation study and example evaluation can be found on GitHub under <https://github.com/daniel-gerhard/RichardsGrowthModel.git>

References

- CLAESKENS, G. & HJORT, N.L. (2008). *Model Selection and Model Averaging*. Cambridge: Cambridge University Press.
- COOMBE, B.G. & MCCARTHY, M. (2000). Dynamics of grape berry growth and physiology of ripening. *Australian Journal of Grape and Wine Research* **6**, 131–135.
- DAI, Z.W., GENARD, M., LI, S.H. & VIVIN, P. (2009). Analyzing the functional association among seed traits, berry growth and chemical composition in Cabernet-Sauvignon berry (*Vitis Vinifera* L.) using a mathematical growth function. *Journal International des Sciences de la Vigne et du Vin* **43**, 35–44.
- ELLIS, R., MOLTCHANOVA, E., GERHARD, D., TROUGHT, M. & YANG, L. (2020). Using Bayesian growth models to predict grape yield. *OENO One* **54**. doi:10.20870/oeno-one.2020.54.3.2972.
- FERNANDES, T.J., PEREIRA, A.A. & MUNIZ, J.A. (2017). Double sigmoidal models describing the growth of coffee berries. *Ciência Rural* **47**, 1–7.
- GELMAN, A., CARLIN, J.B., STERN, H.S., DUNSON, D.B., VEHTARI, A. & RUBIN, D.B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC, 3rd edn. <http://www.stat.columbia.edu/~gelman/book/>.
- GEMAN, S. & GEMAN, D. (1984). Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. *IEEE Transactions on Pattern Analysis and Machine Intelligence PAMI-6*, 721–741.
- GENARD, M., REICH, M., LOBIT, P. & BESSET, J. (1999). Correlations between sugar and acid content and peach growth. *The Journal of Horticultural Science and Biotechnology* **74**, 772–776.
- GREGORCZYK, A. (1998). Richards plant growth model. *Journal of Agronomy and Crop Science* **181**, 243–247.
- HASTINGS, W.K. (1970). Monte Carlo sampling methods using Markov chains and their applications. *Biometrika* **57**, 97–109.
- JORGENSEN, M.A. (1981). Fitting animal growth curves. *The New Zealand Statistician* **16**, 5–15.
- METROPOLIS, N., ROSENBLUTH, A.W., ROSENBLUTH, M.N. & TELLER, A.H. (1953). Equation of state calculations by fast computing machines. *The Journal of Chemical Physics* **21**, 1087.
- RICHARDS, F.J. (1959). A flexible growth curve for empirical use. *Journal of Experimental Botany* **10**, 290–300.
- SEBER, G.A.F. & WILD, C.J. (2003). *Nonlinear Regression*. Hoboken, New Jersey: John Wiley & Sons, Inc.
- TJØRVE, E. & TJØRVE, K.M. (2010). A unified approach to the Richards-model family for use in growth analyses: Why we need only two model forms. *Journal of Theoretical Biology* **267**, 417–425.
- VEHTARI, A., GELMAN, A. & GABRY, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing* **27**, 1413–1432.
- YAO, Y., VEHTARI, A., SIMPSON, D. & GELMAN, A. (2018). Using stacking to average Bayesian predictive distributions (with discussion). *Bayesian Analysis* **13**, 917–1007.